

Joseph Brijs

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EDUCATION

Utah State University

Bachelor of Science: Computer Science

May 2025

GPA: 3.7

EXPERIENCE

Student Software Developer and Research Assistant

Nov 2024 - Present

Utah State University ASPIRE Lab

Logan, Utah

- Developed a Rust program to optimize charger efficiency at a Utah Transit Authority electric bus depot, reducing peak energy demand and expected to lower energy costs by 30%.
- Engineered an EV charger simulator in Rust, enhancing testing workflows and simulation fidelity for both the lab and external partners.
- Performed extensive data analysis and preprocessing to train ML models for predicting in-route electric bus energy consumption.

Software Engineer - Intern

May 2024 – Nov 2024

Elektrik

Lindon, Utah

- Built a Rust/Angular search interface for electrical accessories which contributed to a 25 % increase in revenue.
- Reduced credit application time from days to minutes by creating an adaptive form, automating workflows, generating PDFs, and integrating the process into the web app dashboard
- Aided in the transition from Angular to React, reducing the frontend codebase by 40%, improving UI/UX, and enhancing developer efficiency which resulted in a more maintainable codebase

PROJECTS

Stock Price Neural Network Web App | *Python, PyTorch, Flask, React.js, AWS, Docker*

- Built an automated data pipeline in Python to ingest daily stock prices and engineer features, persisting processed data to Amazon S3 for inference.
- Designed, trained, and evaluated LSTM networks (PyTorch) to forecast next-day prices for major tech equities, achieving an average RMSE of 2.5%.
- Orchestrated serverless inference: scheduled AWS Lambda jobs (via EventBridge) to fetch new data post-market close and run predictions.
- Exposed real-time predictions and historical vs. forecasted comparisons through RESTful API endpoints (Lambda + API Gateway).
- Developed a React.js frontend to visualize predictions and metrics, hosted on S3/CloudFront with DNS via Route 53 (faangfinance.com).

Spotify Genre Clustering (K-Means) | *C++, OpenMP, MPI, CUDA, Python*

- Implemented a serial C++ version of K-Means to cluster 1.2 million Spotify tracks by genre.
- Parallelized K-Means with OpenMP (shared-memory) and MPI (distributed-memory) in C++, ensuring identical results to the serial implementation.
- Developed a Python visualization script to project high-dimensional clusters into a 3D feature space.
- Created a Python validation script to automatically verify that cluster assignments were consistent across all implementations.
- Conducted strong and weak scaling studies on University of Utah CHPC systems

AI Workout-Plan Web App | *React.js, Django, TailwindCSS, Git*

- Built a React.js + Django web app with TailwindCSS to create, store, and track workout plans.
- Connected OpenAI Assistants API to offer both manual and AI-generated workout routines.
- Implemented a dynamic UI for plan creation/editing, enabling drag-and-drop reordering of exercises.

SKILLS

Languages: Rust, JavaScript, Python, Java, C++, HTML/CSS

Frameworks & Libraries & Tools: React.js, Angular, Django, Flask, Redux, Pandas, Numpy, PyTorch, Git, AWS, MongoDB, Docker